

RAJDEEP SINGH PARIHAR

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INTERNSHIP

Biomedical Engineering Intern

During : Jan-2026

Mahaveer Hospital — Indore, Madhya Pradesh

- Gained hands-on exposure to **biomedical equipment operation** and maintenance within a clinical hospital environment.
- Assisted biomedical engineers in **calibration, servicing, and troubleshooting** of diagnostic and therapeutic medical devices.
- Observed and documented workflows for **medical imaging systems** and patient monitoring equipment.
- Developed understanding of **clinical systems integration** and hospital-grade safety standards for electronic equipment.

PROJECTS

RC Drone (Remote-Controlled)

Designed and assembled a fully functional RC drone from scratch, integrating flight controller, coreless motors, motor drivers, and LiPo battery systems.

Programmed flight stabilization and control logic, PWM-based speed control, achieving stable hover and directional control.

RC Car (Remote-Controlled)

Built an RC car using custom circuit design with motor drivers and a radio-frequency remote control module.

Implemented differential steering for precise maneuverability.

Servo Motor Calibrated Robotic Arm

Engineered a multi-axis robotic arm using servo motors calibrated via microcontroller for precise angular positioning.

Developed firmware for sequential movement patterns, demonstrating principles applicable to prosthetic and rehabilitation devices.

IoT Room Temperature & WiFi Measurement System

Developed an IoT-based environmental monitoring system using ESP32, DHT sensors, and a web dashboard.

Streamed live sensor data (temperature, humidity, WiFi signal strength) to a **Web dashboard** for remote monitoring.

WiFi Chatroom over IoT

Built a real-time peer-to-peer chat application hosted on a local ESP32 web server, enabling communication over a shared WiFi network without internet.

Implemented HTTP request handling and a responsive browser-based UI for multi-user messaging.

KEY ACHIEVEMENTS

- **Smart India Hackathon (SIH)** — Selected among Top 50 teams nationally; advanced to the national-level round.
- **MEDHA Hackathon** — **IIT Bombay** — Participated in the MEDHA Hackathon organised by IIT Bombay, competing on healthcare innovation.
- **Resurgence Innovation tech-** Participated in Resurgence hackathon, and got commendation by judges for new ideas.
- **Copyright Registered** — Copyright registered for a software application developed as part of a collaborative team project.
- **NEET Examination** — Scored 99 percentile in Physics and Biology, demonstrating exceptional academic aptitude.

TECHNICAL SKILLS

Electronics & Hardware: Basic Electronics, Circuit Design & Troubleshooting, Biomedical Instrumentation, Servo/Motor Control

IoT & Embedded Systems: ESP32, Arduino, DHT Sensors, PWM Control, Web Server on Microcontroller

Biomedical Knowledge: Medical Imaging Basics, Medical Equipment Basics, Clinical Systems, Rehabilitation Devices

Patent writing: written multiple patents on simple tools as well as complex systems.

Soft Skills: Documentation & Reporting, Problem Solving, Innovation, Team Collaboration, Leadership

EDUCATION

Class 10th and 12th form **Kendriya Vidyalaya, Dwarka sec-8, SPG, New Delhi.**

Lovely Professional University, Punjab

2025 – 2029

Bachelor of Technology (B.Tech) – Biomedical Engineering — First Year | Expected Graduation: 2029